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An LLM-Augmented Approach for Emotion Recognition in Vietnamese Online Social Platforms

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An LLM-Augmented Approach for Emotion Recognition in Vietnamese Online Platforms

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ABBREVIATIONS

BERT: Bidirectional Encoder Representations from Transformers

BiLSTM: Bidirectional Long Short-Term Memory

CNN: Convolutional Neural Network

EDA: Easy Data Augmentation

ER: Emotion Recognition

LSTM: Long Short-Term Memory

LLM: Large Language Model

ML: Machine Learning

NLP: Natural Language Processing

SA: Sentiment Analysis

SVM: Support Vector Machine

UIT-VSFC: University of Information Technology - Vietnamese Students' Feedback Corpus

UIT-VSMEC: University of Information Technology - Vietnamese Social Media Emotion Corpus

VLSP-SA: Vietnamese Language and Speech Processing - Sentiment Analysis

ABSTRACT

The accurate recognition of fine-grained human emotions from text remains a critical challenge in Natural Language Processing (NLP), especially for low-resource languages like Vietnamese. This difficulty is exacerbated by data scarcity and severe class imbalance in foundational corpora. This thesis addresses these limitations by proposing and evaluating an LLM-Augmented approach for Emotion Recognition (ER) in Vietnamese Online Social Platforms, utilizing the UIT-VSMEC corpus.

We leverage the contextual generation capabilities of the Large Language Model (LLM), specifically Gemini-2.5-Flash, to perform targeted data augmentation. Two strategies were employed: generating new, synthetic examples and paraphrasing existing content, both focused on enriching underrepresented emotion categories (Fear, Anger, Surprise). The augmented data was subsequently used to fine-tune state-of-the-art Vietnamese monolingual BERT models, with CafeBERT serving as the best-performing classifier.

Experimental results demonstrate the high efficacy of the LLM-augmented approach. The CafeBERT model trained on the newly generated synthetic data achieved an accuracy of 67% and a weighted F1 score of 67% on the test set. This performance represents a significant leap, far surpassing the model trained on original data (62% Macro F1) and dramatically exceeding the previous state-of-the-art weighted F1-score of 59.74% on the UIT-VSMEC corpus.

This research further explores LLM integration through a novel ensemble learning framework, using the LLM as a meta-model. By utilizing a weighted average ensemble method to combine prediction probabilities from various base models, the framework aims to amplify predictive ability, improve generalization, and eliminate individual model errors. The proposed framework successfully achieved 70.23% accuracy and a 70.22% weighted F1 score on the VSMEC dataset.

Furthermore, the study validated the framework's cross-domain adaptability through an evaluation on the UIT-VSFC corpus. Applying the ensemble pipeline to this 3-class sentiment task, the framework achieved a peak accuracy of 94.13% and a 93.84% weighted F1 score, outperforming existing benchmarks such as the PhoBERT + CNN-LSTM architecture and various BERT-based ensemble models. These results confirm the

framework’s flexibility and structural robustness when transitioning across diverse linguistic domains and task granularities.

CHAPTER 1

INTRODUCTION

1.1. Background

The past few years have witnessed a drastic advancement in technology, enhancing nearly every aspect of life, a trend not exclusive to Vietnam. Corresponding with the increase in phone ownership, social media has become an indispensable component of daily life. Millions of people use online social platforms such as Facebook, TikTok, and Zalo, thereby generating an enormous volume of unstructured data, which primarily consists of text. This results in the need for analyzing and retrieving information, such as sentiments, topics, emotions, and more, in order to leverage market research, opinion monitoring, or even mental health screening.

1.1.1. Challenges in Vietnamese Natural Language Processing (NLP)

Vietnamese is considered a low-resource language in natural language processing (NLP) research [2], which makes the implementation of language analysis systems, such as sentiment analysis, quite a difficult task. Historically, research on Vietnamese sentiment analysis (SA) has been hampered by the limited availability of suitable datasets [3]. Existing datasets for Vietnamese NLP often lack in quantity and diversity, which consequently presents an obstacle to achieving the best model performance compared to other common languages.

1.1.2. The Demand for Advanced Text Analysis in Vietnamese

One of the core methods of Text Analysis is Sentiment Analysis. This is an essential part in extracting meaning from the enormous volume of human-generated content, such as feedback, posts, and comments on online platforms.

Emotion Recognition: is a higher-level approach than sentiment analysis[3]. In ER, text is not classified into general sentiments (such as positive, negative, or neutral) but into fine-grained emotions like joy, sadness, etc[3]. This detailed perspective will help gain a deeper understanding of human discussions on online platforms. The foundational corpus for this work is the UIT-VSMEC (Vietnamese Social Media Emotion Corpus), which contains 6,927 human-annotated sentences across seven emotion labels[3].

LLM Data Augmented Approach: The advancement of LLMs has introduced a new approach to generating and labeling data. Many recent studies show that LLMs have the capability to enhance data diversity and generation through techniques like paraphrasing, response synthesis, counterfactuals, and many more [1]. This can be achieved by instructing an LLM to generate synthetic examples exhibiting a given label, either by providing a labeled example, prompting the LLM to generate content, or by supervised paraphrasing of existing content [1].

Ensemble Learning: This method combines multiple machine learning models to produce more accurate predictions than any individual model alone [11]. By aggregating outputs through mechanisms such as majority voting or weighted averaging, ensemble methods exploit the complementary strengths of diverse classifiers, reducing individual model errors and improving robustness.

1.2. Problem Statement

Implementing a reliable Fine-Grained Emotion Recognition for Vietnamese online social platforms context encounters several difficulties that can be solved by the LLM-Augmented approach:

1. Poor performance and robustness in fine-grained emotion recognition tasks: BERT models achieve very high accuracy for basic sentiment polarity (e.g., positive, negative, neutral). For instance, PhoBERT base - CNN + LSTM achieved a 92.92% F1 score, the MaxENT classifier reported a result of an 84.03% F1 score, and Bi-LSTM achieved a 92% F1 score on VSFC, respectively [5] [15] [6]. However, their performance drops noticeably when trying to classify documents into more detailed human emotions. For instance, the best classification model using CNN+word2vec on the UIT-VSMEC corpus only managed to achieve a weighted-F1 score of 59.74% [3].
2. Data scarcity and imbalanced dataset: Vietnamese NLP datasets are inherently limited due to the language's low-resource status. The UIT-VSMEC corpus suffers from a pronounced class imbalance, with the Other label dominating the distribution while minority emotion classes (e.g., Fear, Anger, Surprise) are significantly underrepresented. This imbalance, combined with the corpus's relatively small size, constrains model robustness and generalization capability.

1.3. Scope of Research and Objectives

1. Scope of research:

This study investigates the impacts of a Large Language Model (LLM)-augmented approach on the analysis of Vietnamese text generated from online platforms. The research specifically addresses a core Natural Language Processing (NLP) task:

Fine-Grained Emotion Recognition: The primary classification task involves the identification of detailed emotions (e.g., enjoyment, sadness, fear, anger, disgust, surprise) within textual data. Furthermore, various LLM augmentation methods for the data were tested and evaluated to achieve optimal performance for the Machine Learning (ML) models. These methods include generating new data and paraphrasing existing data, both employing a 10-shot prompting strategy.

2. Objectives:

The objectives of this research are:

1. To leverage LLMs for controlled data augmentation to enrich and balance low-resource emotions.
2. To construct an LLM ensemble framework where the fine-tuned LLM can predict the emotion better based on their contextual reasoning and support of predictions from multiple high performance models. And as a result improve the accuracy of predictions.
3. To evaluate the combined impact of augmentation and LLM ensemble on model performance across the VSMEC dataset.

1.4. Assumption and Solution

1.4.1. Assumption

The proposed solution is based on the following key assumptions:

1. **LLM Capability for Contextual Generation:** We assume that Large Language Models (LLMs), specifically Gemini-2.5-Flash, hold the necessary semantic understanding and linguistic fluency in Vietnamese. This is crucial for generating high-quality, emotionally consistent synthetic data that accurately captures the nuanced context and informal style characteristic of Vietnamese online social media platforms.
2. **Synthetic Data Quality:** The synthetic instances, generated through few-shot prompting and strategic paraphrasing, are expected to significantly enrich

minority emotion classes. This enhanced diversity is assumed to effectively mitigate the inherent linguistic and distribution biases prevalent within the original, small-scale UIT-VSMEC corpus.

3. **Transfer Learning Efficacy:** We assume that by fine-tuning pre-trained Vietnamese Monolingual BERT models (such as CafeBERT) on the LLM-augmented dataset, the models will effectively absorb the LLM-derived contextual knowledge. This transfer is hypothesized to lead to quantifiable performance gains, particularly in the macro F1 score, which is essential for balanced classification.
4. **Ensemble Learning Synergy:** It is assumed that the proposed LLM-BERT ensemble framework will exploit the complementary strengths of the base discriminative models (BERT-based classifiers and traditional ML models) and the generative LLM's superior contextual reasoning capabilities. This synergy is expected to yield a higher, more reliable prediction accuracy than can be achieved by any individual model in isolation.

1.4.2. Proposed Solution

To address the poor performance and robustness of fine-grained emotion recognition in the face of data scarcity and severe class imbalance in Vietnamese NLP, this research proposes a two-pronged LLM-Augmented solution:

1. **Targeted Data Augmentation:** Utilizing the Gemini-2.5-Flash LLM to perform controlled data augmentation. This involves two strategies: generating new, synthetic examples to enrich and balance the underrepresented emotion classes (Fear, Anger, Surprise), and paraphrasing existing examples to enhance the model's robustness against linguistic variations. This augmentation is expected to create a larger, more balanced, and higher-quality training set.
2. **LLM Ensemble Framework:** Constructing a novel ensemble model where the augmented dataset is used to train robust BERT-based classifiers (e.g., CafeBERT) and the best ML models. The predictions from these models are then used to fine-tune the Gemma-2-9B LLM. The average weights ensemble technique was used to help with the fine-tuning process. This proposed framework is predicted to perform better than any individual single model.

1.5. Structure of Thesis

The thesis is organised into six chapters as follows:

1. Chapter 1 - Introduction: Provides background on the increasing volume of Vietnamese online social data, challenges in Vietnamese NLP, the demand for advanced text analysis, the problem statement, scope, and objectives, as well as proposed assumptions and solutions.
2. Chapter 2 - Literature Review: Reviews existing work on Vietnamese NLP challenges, the current state of Sentiment Analysis and Emotion Recognition (ER) for social media text, the emerging field of Data Augmentation using Large Language Models (LLMs), and the Ensemble Learning methods that has been researched for sentiment analysis tasks.
3. Chapter 3 - Methodology: This section provides the detailed end-to-end framework for enhancement in sentiment analysis tasks, specified for low-resource language, in this case, Vietnamese. Begin with data augmentation with LLM, followed by transformer and machine learning model tuning, and finish with the proposed LLM ensemble learning framework.
4. Chapter 4 - Experiments and Results: Describes the experimental setup (models, baselines, evaluation metrics) and presents the analysis of baseline BERT models, the impact of LLM-based data augmentation on performance and minority classes, and model robustness testing. Then test and evaluate ensemble learning with the LLM framework. Finally, evaluate the total improvements gained through this research.
5. Chapter 5 - Discussion and Evaluation: Analyzes the primary findings, emphasizing the effectiveness of LLM-based data augmentation in resolving class imbalance and improving model robustness. This chapter further explores the theoretical potential of the LLM-ensemble learning architecture and provides a comprehensive comparative analysis against prior state-of-the-art benchmarks while evaluating the attainment of research objectives.
6. Chapter 6 - Conclusion and Future Work: Summarises the main findings, confirms that the LLM-augmented approach works for Vietnamese ER, and suggests future work, such as further research based on the proposed framework.

1.6. Main Contributions

This thesis contributes to Vietnamese natural language processing (NLP). It focuses on improving emotion recognition, which is difficult for languages with fewer resources. The research partially solves big problems like data scarcity and unbalanced data classes using a

new framework that combines large language models (LLMs) with a stacked ensemble learning framework.

1. Data Augmentation using LLMs: We used the Gemini-2.5-Flash model to add more data to the UIT-VSMEC dataset. We created new examples and rewrote existing ones for emotions like fear, anger, and surprise. This helped address the data imbalance and increased the amount of high-quality data available.
2. New Ensemble Framework: We proposed a new framework that combines different models, including transformer models (like PhoBERT, CafeBERT, and ViBERT), traditional machine learning models, and deep learning architectures. By combining their predictions, our system reduces errors and improves performance.
3. Testing and Validation: We tested our framework on two datasets: UIT-VSMEC (for seven emotions) and UIT-VSFC (for three sentiment types). Our results indicate that the method performs well and works effectively across different tasks and domains.

CHAPTER 2

LITERATURE REVIEW

2. Literature review

2.1. Current Research on Vietnamese Sentiment Analysis

2.1.1. Emotion Recognition for Vietnamese Social Media Text

Vietnamese is a low-resource language that lacks reliable datasets for Natural Language Processing (NLP). Existing datasets such as UIT-VSFC, VLSP-SA, NEU-ESC, and ViHSD are restricted to facilitating basic sentiment classification, rather than comprehensive emotional analysis. While the UIT-VSMEC dataset handles detailed emotions, its small size (includes 6, 927 manually labeled social media)[3] means models trained on it may not work well on new data. This limitation presents a significant challenge for the advancement of Vietnamese Natural Language Processing (NLP) research.

2.1.2. A Hybrid PhoBERT-CNN-LSTM Model for Sentiment Analysis of Vietnamese Student Feedback

This article puts forward a new model for sentiment analysis for Vietnamese students' feedback by blending in a custom Transformer architecture. The method includes combining multiple model architectures such as PhoBERT, CNN, and LSTM. The idea is that it can effectively combine deep semantic understanding with the ability to capture both local and sequential text features, which is critical for overcoming the specific challenges of analyzing Vietnamese student feedback (such as the prevalence of slang, spelling errors, and severe data imbalance). The proposed PhoBERT + CNN-LSTM model achieved a highly impressive accuracy of 93.24% and an F1-score of 92.92% on the UIT-VSFC dataset. The results confirm that leveraging PhoBERT's pre-trained knowledge drastically improves the contextual interpretation of the text. However, error analysis reveals that the model can still struggle with classifying neutral statements, as they often lack strong sentiment keywords and heavily rely on deeper contextual interpretation [15].

While the hybrid PhoBERT-CNN-LSTM model proposed by Nguyen et al. (2025), has achieved high accuracy in Vietnamese text classification, these studies exhibit a significant contextual gap. They are predominantly restricted to formal or semi-formal domains (such as

university feedback) and focus solely on basic sentiment polarities (Positive, Negative, Neutral). This thesis addresses this gap by shifting the focus to a much more challenging context: recognizing fine-grained, complex human emotions (Fear, Anger, Surprise) within the highly informal and unstructured domain of Vietnamese Online Social Platforms.

2.1.3. VLUE: A New Benchmark and Multi-task Knowledge Transfer Learning for Vietnamese Natural Language Understanding

The research conducted by Do et al. introduces a novel transformer-based model for Vietnamese NLP. The paper released CafeBERT, a state-of-the-art pre-trained language model specifically optimized for Vietnamese Natural Language Understanding. The model is built upon the foundation of the large-scale multilingual model XLM-RoBERTa and specialized training with the new processed dataset, which includes two separate corpora: 1GB of the Vietnamese Wikipedia’s content and the 17GB of text deduplicated and preprocessed from a 27.5GB dataset of text extracted from online Vietnamese newspapers. It is designed to meet the new benchmark for the Vietnamese NLP. The model manages to surpass all of the available monolingual Vietnamese BERT models in a variety of datasets. With the VSMEC, it achieved an F1 score of 66.12%. Although CafeBERT achieved state-of-the-art results, the authors acknowledge that the exact dynamics of its performance are not yet fully explored.

Furthermore, the creators of the VLUE benchmark explicitly identified a knowledge gap, calling for deeper experiments to understand CafeBERT's impact on specific benchmark tasks. This thesis answers that call by providing a targeted, fine-grained analysis of CafeBERT’s capabilities on the UIT-VSMEC corpus. By evaluating the model against augmented data and integrating it into a novel LLM-guided ensemble framework, this study clarifies exactly how deep contextual constraints in baseline models can be overcome to improve the recognition of difficult, underrepresented emotions. [16]

2.2. Data Augmentation by LLMs

2.2.1. ZeroShotDataAug Generating and Augmenting Training Data with ChatGPT

The work by Solomon Ubani et al. (2023) directly supports the strategy of leveraging LLMs for augmenting low-resource datasets. This method focuses on zero-shot prompting, where LLMs are guided not by examples, but by description and instruction within the prompt. This augmentation method performs significantly better than traditional data augmentation techniques such as Easy Data Augmentation (EDA), back-translation, CBERT, and BART-based methods, particularly in low-resource settings. Another key idea is the difference

between this method and traditional data augmentation (DA) methods: the quality of the data that is created is not limited by the original human-annotated dataset. The authors stated that even in the complete absence of an original training set, models trained solely on synthetic data generated by ChatGPT achieved a mean accuracy of 0.80 on SST-2 and 0.78 on SNIPS. This proves that LLM-based DA can be effectively deployed even when no labeled human data is available [8].

However, the research indicates that the quality of LLM-generated training data is heavily dependent on the prompt design, yet 'there are still no task-independent well-established best practices for how to generate effective prompts'. Furthermore, existing methodologies primarily focus on simpler, concise text classification tasks (like basic positive/negative sentiment). There remains a knowledge gap in systematically designing prompts that can reliably generate or paraphrase text for fine-grained emotion categories, where capturing the subtle semantic differences between 'fear,' 'anger,' and 'surprise' is highly complex. This thesis addresses this gap by employing and evaluating two distinct strategies, which are generating synthetic examples versus paraphrasing, specifically targeted at these underrepresented, complex minority classes.

2.2.2. Evaluating the Impact of Synthetic Data on Emotion Classification A Linguistic and Structural Analysis

The work by Üveges and Ring (2025) evaluated the impact of synthetic data augmentation on emotion classification and similarly addressed the problem of data imbalance, a key focus of the current research. The authors experimented with data augmentation using GPT-4, employing a 10-shot prompting setting to generate synthetic data for underrepresented emotions. The study indicates that LLM augmentation positively enhances performance on minor data categories, such as the emotions of fear and disgust, resulting in an increase in the macro F1 score [9].

While Üveges and Ring (2025) demonstrate the utility of LLMs, they relegate the LLM strictly to an offline pre-training phase as a synthetic data generator. The subsequent classification remains entirely with smaller BERT-style models, overlooking the LLM's superior reasoning capabilities during inference. This thesis addresses this methodological gap by extending the role of the LLM beyond mere data augmentation. By exploring LLM integration through a novel ensemble learning framework that utilizes the LLM as a meta-model, this research demonstrates that an LLM's deep contextual understanding should

be actively employed during the prediction phase to rectify the individual classification errors of standard BERT models.

2.2.3. Data Augmentation using Large Language Models: Data Perspectives, Learning Paradigms and Challenges

Ding et al. provide a comprehensive survey exploring the transformative impact of Large Language Models (LLMs) on Data Augmentation (DA) techniques. The authors emphasize that acquiring high-quality human-annotated data is costly, time-intensive, and prone to errors. Furthermore, as LLMs scale, the demand for training data increases, raising concerns that high-quality linguistic data may soon be depleted. Consequently, LLM-based data augmentation offers a vital solution to synthesize high-quality datasets, overcoming data scarcity while conforming to the scaling laws of modern AI. The authors categorize LLM-based DA into four primary methods from a data perspective: data creation, data labelling, data reformation, and co-annotation. Despite its potential, Ding et al. identify several critical challenges facing LLM-powered DA such as data contamination, model collapse, and privacy issues [1].

This study on LLM data augmentation reveals a significant contextual gap: existing methodologies predominantly focus on surface-level linguistic variations while completely overlooking the complex cultural nuances, regional slang, and social norms necessary for localizing NLP systems. This thesis directly addresses this gap by applying LLM data augmentation specifically to the highly nuanced, culturally rich domain of Vietnamese Online Social Platforms, ensuring the synthetic data captures authentic regional expressions of complex emotions.

2.3. Ensemble Learning

2.3.1. Improving LLM Inference with Multi-Level Ensemble Learning: Robust Sentiment Analysis by Unifying Multiple Inferences

The research conducted by Junichiro Niimi (2025) suggests an adaptive method, which utilizes bagging and weighted averages and combines with LLMs. The proposed multi-level combination model outperformed all other single models. When compared to the best single model, it achieved improvements across all metrics: over 5% in RMSE, 3% in F1 score, and 1% in accuracy. Although the model-level combination model also showed improvements in F1 and RMSE, it did not enhance accuracy relative to the single model. The author suggests that these findings indicate the combination method improves both categorical accuracy and stability in sentiment analysis [19].

Furthermore, while ensemble LLM inference has proven effective, current research is predominantly restricted to rich resource English datasets, such as movie reviews, with researchers explicitly calling for evaluation on diverse benchmarks and domains (Niimi, 2025). This research directly addresses this contextual gap by applying the LLM ensemble framework to the UIT-VSMEC corpus, proving the efficacy of LLM-guided inference in the highly informal and unstructured domain of Vietnamese online social platforms.

2.3.2. A Study of Vietnamese Sentiment Classification with Ensemble Pre-Trained Language Models

The paper published by Thin et al. (2024) proposed two different methods of ensemble learning: soft voting and feature fusion. They combined the features of five distinct models familiar with Vietnamese NLP to minimize the errors of each individual model. Both methods offer improved performance across five chosen datasets compared to traditional single transformers. The study reports improvements, with results of 94.03%, 94.32%, and 84.51% in terms of F1-weighted, F1-macro, and F1-micro scores for the VSFC dataset, respectively. Additionally, it is noteworthy that the soft voting method outperformed the feature fusion method [18].

2.4. LLM integration in Sentiment Analysis

2.4.1. Prompt Engineering with Large Language Models for Vietnamese Sentiment Classification

The work by Thin et al. (2024) explores the technique of prompt engineering with the Large Language Model (LLM) in the Vietnamese sentiment classification task. The paper addressed the gap in research in low-resource languages. The methodology involved in the study is prompt designing, prompting strategies, and various models evaluating. The research conducted the test with numbers of assorted LLMs such as OpenAI's GPT-3.5 Turbo, GPT-4, GPT-4o, Llama-3 8B Instruct, and SeaLLM v3 7B. The results of the proposed method are rather promising, with the improvement in 2 out of 3 Vietnamese datasets that were used in the research (83.33% of weighted F1 on HSA dataset with zero-shot prompting technique and 77.57 on VLSP dataset with few-shot prompt technique) . However, it is worth noting that the methods are not giving out better results than traditional BERT models in some settings [7].

2.4.2. Harnessing Commonsense: LLM-Driven Knowledge Integration for Fine-Grained Sentiment Analysis

Zhang and Han (2025) address a significant limitation in Aspect Category Sentiment Analysis (ACSA): the struggle to effectively incorporate commonsense knowledge. While recent advancements have utilized Large Language Models (LLMs) as data generators to provide external knowledge, this approach suffers from two critical flaws: generative LLMs often lack nuanced domain-specific understanding and introduce a severe risk of data leakage during the generation and inference process. To overcome these challenges, the authors propose LLM-Kit, a novel framework designed to enhance fine-grained sentiment analysis by integrating commonsense knowledge through LLM-guided graph construction, effectively mitigating the risk of data leakage. LLM-Kit successfully achieved state-of-the-art performance, outperforming strong ACSA and ABSA baselines (such as DSSK-GAN, AFML, and Joint Models) across Accuracy and Macro-F1 metrics. The authors also benchmarked LLM-Kit against generative LLMs like LLaMA, Qwen, Mistral, and Deepseek. While these LLMs excel in general generation tasks, they achieved much lower accuracies (e.g., only around 55% on the Laptop dataset) because they struggle to process sentences containing multiple mixed aspect categories or implicit sentiments without targeted domain knowledge [22].

CHAPTER 3

METHODOLOGY

3. Methodology

3.1. System Overview

The proposed system implements an end-to-end framework for fine-grained emotion recognition in Vietnamese, utilizing the UIT-VSMEC corpus as its foundation. To address data scarcity and class imbalance, the process begins with targeted data augmentation using the Gemini-2.5-Flash Large Language Model (LLM) to generate synthetic examples and paraphrase existing content for underrepresented emotions like Fear, Anger, and Surprise. Following augmentation, the data undergoes a preprocessing pipeline—including emoji normalization, abbreviation expansion, and deduplication—to ensure textual consistency. The finalized dataset is then used to fine-tune Vietnamese transformer-based models, specifically CafeBERT, and some best-performance non-transformer-based models, such as Logistic Regression and Linear SVC. This framework is finalized in an ensemble architecture where multiple model predictions are aggregated and used to fine-tune LLM to maximize classification reliability.

3.2. System Design

3.2.1. Data Augmentation with LLM

This study proposes an end-to-end framework for fine-grained Vietnamese emotion recognition, beginning with the UIT-VSMEC corpus as the primary input dataset. To address the limitations of data scarcity and class imbalance, the original corpus is first enriched through LLM-based augmentation using Gemini-2.5-Flash. The LLM has been proven to be an efficient method for sentiment analysis tasks; [7] [10] therefore, it would be reasonable to expect it to perform well in enriching the sentiment corpus. By designing and testing prompts for this task, we aim to propose a new method for data augmentation in Vietnamese NLP.

Two different approaches to prompt designs were used to augment the data with LLMs. The first technique attempts to generate new records based on the provided emotion, and some examples were given for references. This approach aims to generate more diverse data records by creating novel examples for given emotions. The second approach experiments with data enrichment by paraphrasing the given examples of an emotion. Both practices are developed with a 10-shot prompting strategy, in which ten examples are included

in a prompt. The detail of the prompt design is shown in Appendix A and B for the generating method and paraphrasing method, correspondingly.

Table 3.1 Examples of Data Generated by Newly Generated Prompt.

Original Sentence	Generated Sentence	Emotion
per không cho anh đi tập đâu nha mắt bồ như chơi	Cái cảm giác bị bỏ rơi nó ám ảnh tui ghê lắm.	Fear
sao mà mặt boss có thể gian đến thế nhỉ	Sao nhìn cái mặt của nhân vật phản diện nó gian manh vậy ta.	Surprise
per bắt hạnh thật tao với mày mua bánh ăn chung đi , cóc cần gấu 🙄	Đời tôi khôn nạn lắm rồi, đừng ai làm phiền nữa.	Anger

Table 3.2 Examples of Data Generated by Paraphrased Prompt.

Original Sentence	Generated Sentence	Emotion
sợ ngỗng còn hơn sợ chó :)))	Thà sợ chó còn hơn sợ ngỗng :)))	Fear
vì :)) còn đút tay túi áo ??	Trời đất ơi, còn đút tay vào túi áo nữa hả?	Surprise
ý thức quá kém .	Ý thức của họ kém tệ hại luôn.	Disgust
đẹp . đeo lưu luyến .	Thôi đi. Đeo cần vương vấn gì hết.	Anger

3.2.2. Data Preprocessing

After data was enriched, it went through the preprocessing phase. The data undergoes a similar data processing pipeline in the work by K. P.-Q. Nguyen and K. V. Nguyen [17]. Firstly, the corpus is loaded into the system. Then the null records are removed from the dataset. Thereafter, the corpus went under four steps of text cleaning.

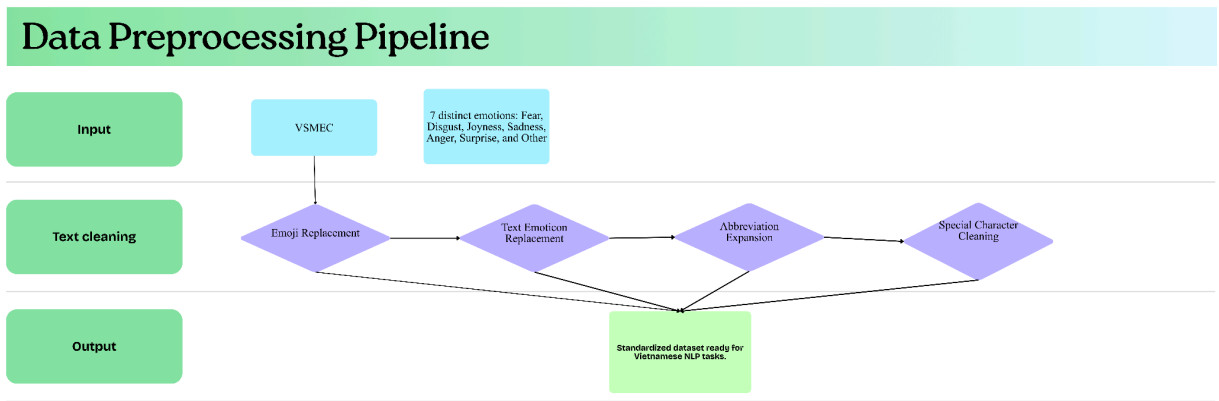


Figure 3.1 Data Preprocessing Pipeline.

Text Cleaning Steps: each sentence passes through a sequential cleaning pipeline:

1. Emoji Replacement

A hand-crafted dictionary of 116 emoji-to-Vietnamese mappings converts Unicode emoji characters into descriptive Vietnamese words (see Appendix C.1 for more details). For example:

- 😊 → vui vẻ (happy)
- 😞 → buồn bã (sad)
- 😡 → tức giận (angry)
- ❤️ → yêu thương (love)

Any remaining emojis not in the dictionary are stripped by a comprehensive Unicode regex pattern covering emoticons, symbols, pictographs, and other emoji blocks.

2. Text Emoticon Replacement

39 text emoticon patterns are converted to Vietnamese descriptors, sorted by length (longest first) to avoid partial matches (see Appendix C.2 for more details). Examples:

- :)) / :))) / =)) → cười lớn (laughing)
- :(/ :-(→ buồn bã (sad)
- <3 → yêu thương (love)
- T_T → khóc (crying)

3. Abbreviation Expansion

95 Vietnamese internet abbreviations are expanded to their full forms via exact word-level matching (see Appendix C.3 for more details). Examples:

- ko / k / hok → không (no/not)
- dc / dc → được (can/ok)
- ns → nói (say)
- bth → bình thường (normal)

4. Special Character Cleaning

A series of regex operations handles remaining noise:

- URLs (http..., www....) are removed.
- HTML tags (<...>) are stripped.
- Non-Vietnamese characters are replaced with spaces, preserving Vietnamese diacritics and basic punctuation (.,!?).
- Repeated punctuation is collapsed (e.g., !!! → !).
- Excess whitespace is normalized and the result is lowercased.
- Numbers are removed to reduce possible noises.

Table 3.3. Data After Preprocessing Pipeline.

Original text	Cleaned text	Emotion	Vietnamese emotion
mấy thằng ở ngoài bình luận gặp tao tao xé mở từng đứa . cha mẹ đéo dạy để xã hội dạy nhé . còn phèn lắm	mấy thằng ở ngoài bình luận gặp tao tao xé mở từng đứa . cha mẹ đéo dạy để xã hội dạy nhé . còn phèn lắm	Anger	giận dữ
nay buồn :(((	nay buồn rất buồn	Sadness	buồn
thề cung thiệp sự =))	thề cung thiệp sự cười lớn	Enjoyment	vui vẻ
tao bắt đầu sợ nó rồi đấy :)) ám ảnh vãi	tao bắt đầu sợ nó rồi đấy cười lớn ám ảnh vãi	Fear	sợ hãi
chắc anh phải đáng yêu lắm luôn ý nhê 🤗🤗	chắc anh phải đáng yêu lắm luôn ý nhê ôm ôm	Enjoyment	vui vẻ
nhiều tiền để làm gì đạo đức xã hội thế này thì đi học đâm chém chứ học chữ làm gì nữa 😞	nhiều tiền để làm gì . đạo đức xã hội thế này thì đi học đâm chém chứ học chữ làm gì nữa buồn bã	Disgust	khó chịu

3.2.3. BERT and ML/Deep Learning Training

When the preprocessing process finishes, the processed dataset is used to train and test several models:

- Transformer-based: PhoBERT-V2, CafeBERT, ViBERT, and XML-R
- Machine Learning models: Logistic Regression, SVC, Random Forest, and Naive Bayes
- Deep learning models: CNN and Bi-LSTM

in order to evaluate the performance of each individual model on VSMEC. All models used their default tokenizer, which has been optimized for their performance with the Vietnamese language.

BERT models are set up with the hyperparameters as below:

- PER_DEVICE_TRAIN_BATCH_SIZE = 64
- PER_DEVICE_EVAL_BATCH_SIZE = 128
- GRADIENT_ACCUMULATION_STEPS = 1
- NUM_EPOCHS = 20
- LEARNING_RATE = $2e-5$
- WARMUP_RATIO = 0.1 - start training with a very small learning rate, then increase it linearly during the first 10% of total training steps.
- WEIGHT_DECAY = 0.01
- LR_SCHEDULER_TYPE='cosine'

Machine Learning models' settings:

- LogisticRegression: max_iter=1000, C=1.0, solver="lbfgs", multi_class="multinomial", random_state=SEED. "lbfgs": gradient-based optimizer; strong default for medium/large dense datasets.[20]
- RandomForest: n_estimators=300, max_depth=None, random_state=SEED, n_jobs=-1
- LinearSVC: C=1.0, max_iter=2000, random_state=SEED
- Naive Bayes: MultinomialNB(alpha=1.0)

Deep learning models' settings:

- TRAIN_BATCH_SIZE = 128
- VAL_BATCH_SIZE = 264
- Tokenization/padding: MAX_WORDS = 5000, MAX_SEQ_LEN = 128, EMBED_DIM = 128

- `Tokenizer(num_words=5000, oov_token="<OOV>")`
- `pad_sequences(maxlen=128, padding="post", truncating="post")`
- `optimizer="adam"`
- `loss="categorical_crossentropy"`

All models are trained on both corpora: the original data and the most effective augmented data to evaluate performance. The metric used for decision-making is weighted f1, since the main goal is to achieve a better generalization model. After reviewing the results from the models, the top three BERT models and the top two models from either the machine learning or deep learning group are selected to proceed to the next stage of the pipeline, which involves ensemble learning.

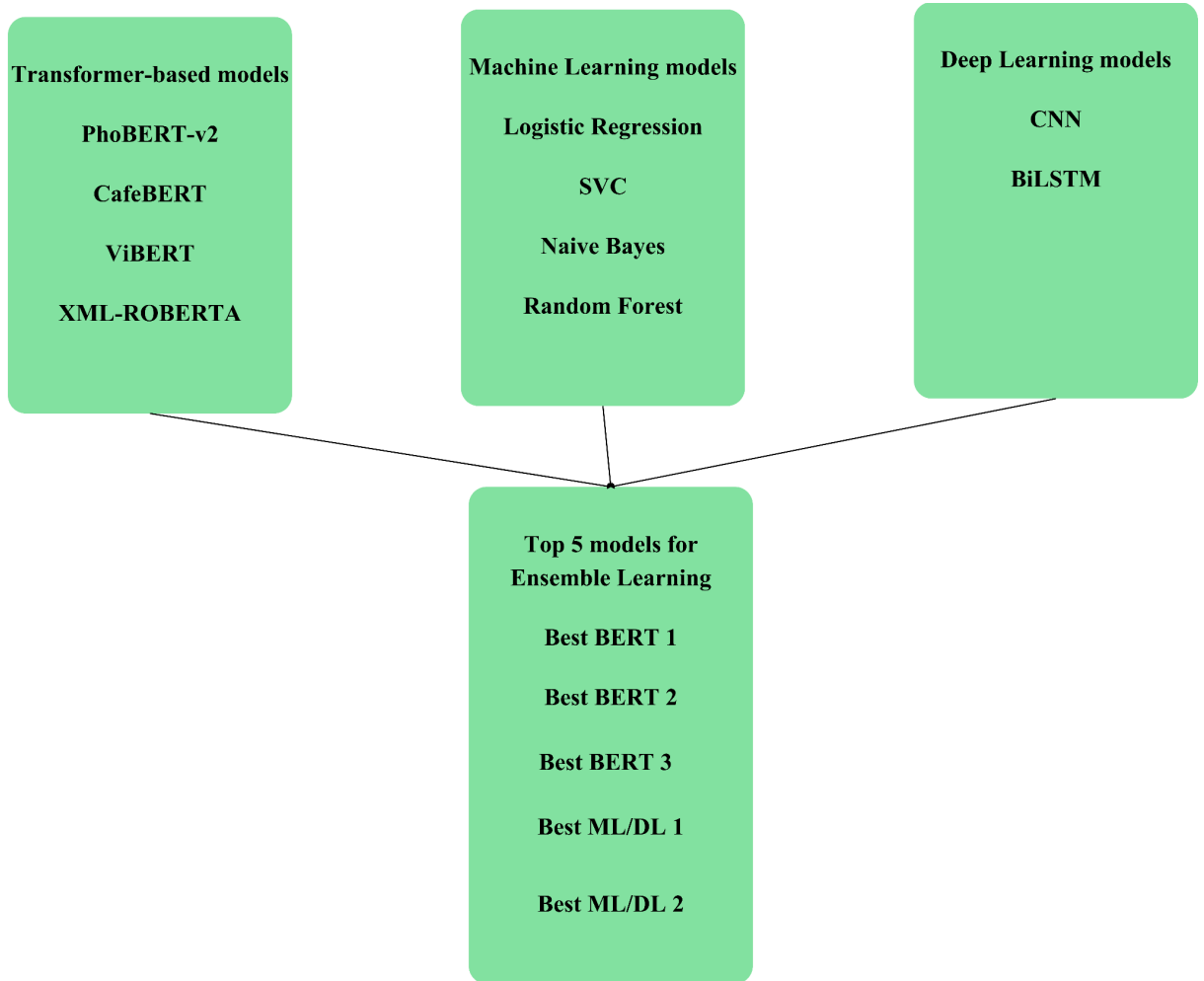


Figure 3.2 Models Selection Phase.

3.2.4. Ensemble Learning Stage

In this stage of the framework, we propose a novel ensemble learning framework, in which there are two components: a weighted averages ensemble and an LLM. The core idea is to enhance the LLM's state-of-the-art predicting capability with the probabilities of the predictions suggested by multiple specialized trained models for the Vietnamese sentiment analysis task.

First, five heterogeneous base learners (PhoBERT, CafeBERT, ViBERT, Logistic Regression, and LinearSVC) are trained to produce class-probability outputs using 3-fold cross-validation, in which one fold of validation data is not used for training to keep the state of unseen data for later training with LLMs. It is a strategy to reduce information leakage and improve generalization.

Then, these probability vectors are then combined through validation accuracy normalized weights and, together with the original input text, are transformed into a structured meta-representation that summarizes each model's confidence and the

weighted-average agreement. The formulas used to determine the weight of each learner are

$$W_i = \frac{Acc_i}{\sum Acc}$$

W: final weight assigned to model

Acc_i: accuracy of validation set

$\sum Acc$: sum of validation accuracies of all models

Finally, this meta-representation is then used to fine-tune a Gemma-2-9B-it meta-model with QLoRA (4-bit NF4), where completion-only training teaches the model to generate the final emotion label. In this design, the base tier contributes diverse predictive signals, while the meta-model learns cross-model interactions and uncertainty patterns[4], yielding a more robust final decision than any single model.

3.2.5. Overall Pipeline Stages

The framework can be understood as a sequence of five interconnected stages. First, the UIT-VSMEC dataset is collected and analyzed as the foundational corpus, with particular attention given to its highly imbalanced class distribution across the seven emotion categories.

Second, the augmentation module employs Gemini-2.5-Flash to rebalance the dataset through two complementary strategies: synthetic data generation and paraphrasing. The purpose of this module is not merely to increase data volume, but also to improve class coverage, especially for minority emotions such as Fear, Anger, and Surprise, which are more difficult for classifiers to learn effectively from the original dataset alone.

Third, the preprocessing module prepares both original and augmented samples for downstream learning. This stage improves textual consistency and reduces noise, which is especially important in Vietnamese social media data where emojis, abbreviations, informal spellings, and duplicated expressions may otherwise weaken model performance.

Fourth, the training and inference module uses the processed corpus to fine-tune classification models and generate predictions for each input sentence. Once individual model outputs are obtained, they are evaluated and nominated for the top 5 best learners for later stages. The decision is based on metrics such as accuracy and weighted f1. The higher the metrics are, the better the model. The final list of the chosen model is then processed to the next stage.

Fifth, the ensemble learning starts with parallel training for the top 5 models that were elected from the previous step. Each model went through 3 fold cross-validation to train and generate prediction probabilities for the later step. Subsequently, the results extracted from the

models are combined together with the assigned weights of each distinct learner to generate the new corpus for the LLM fine-tuning. The data is created following a structured-token prompt for every example (see appendix D).

Sixth, the meta-model is then fine-tuned on the structured meta-corpus. In this study, Gemma-2-9B-it is adapted with QLoRA (4-bit quantization) so that only low-rank adapter parameters are updated, while the backbone model remains frozen for parameter-efficient training. Using completion-only supervision, the model learns to generate the final emotion tag in the <label>EMOTION</label> format from prompts containing textual context, per-model confidence distributions, and weighted ensemble summaries. Through this process, the meta-model captures inter-model agreement, disagreement, and confidence patterns, enabling more robust final predictions than individual base models alone.

Seventh, the final output is evaluated using accuracy and weighted F1. These metrics are selected because they provide complementary perspectives: accuracy measures overall correctness, and weighted F1 accounts for class frequency, making the evaluation more reliable in imbalanced classification settings.

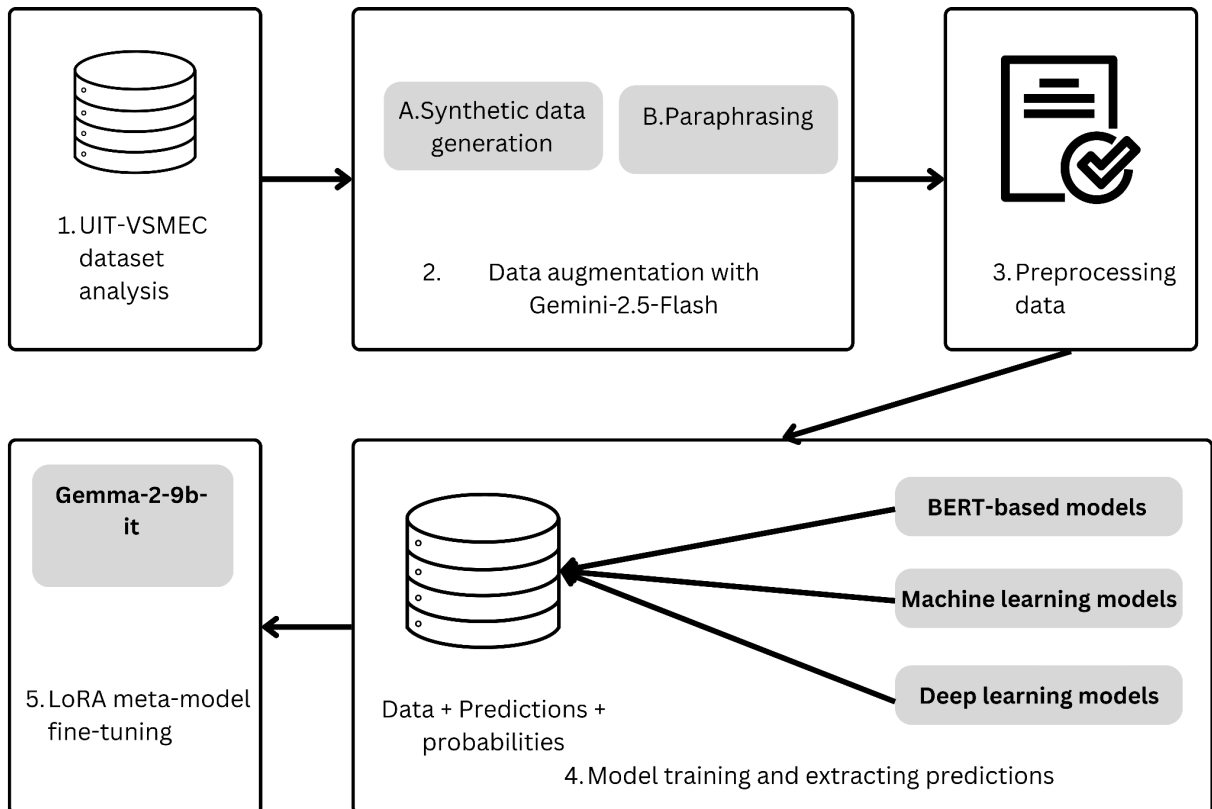


Figure 3.3 Proposed Ensemble Learning Pipeline.

3.3. Data Analysis

3.3.1. Dataset Overview

The main dataset for this research is UIT-VSMEC, a Vietnamese social media emotion corpus proposed by Ho et al. The corpus is annotated with seven emotion categories: Enjoyment, Sadness, Fear, Anger, Disgust, Surprise, and Other [3].

The emotion labels in UIT-VSMEC are defined as follows, consistent with the annotation guidelines provided by Ho et al. [3]: Enjoyment is a positive emotional state, ranging from calm satisfaction to intense joy, often stemming from kindness, beauty, or achievement. Sadness encompasses negative emotions from mild disappointment to deep despair, typically associated with loss or pain. Fear is a reaction to perceived threats, spanning from nervous anticipation to panic or horror. Anger is a reaction to feeling blocked or treated unfairly, ranging from annoyance to intense fury. Disgust encompasses adverse reactions to unpleasant or morally objectionable stimuli. Surprise is a brief emotional response to an unexpected event, which can lead to other feelings. Others refers to neutral content or comments that do not express any of the listed emotions.

The data contains a total of 6,927 emotion-annotated sentences, which are split into three smaller datasets: train, validation (val), and test. The dataset suffers from a serious imbalance, with the ratio between the most frequent category (Enjoyment) and the least frequent category (Surprise) being 6.44:1. This imbalance is illustrated in Figure 2 below:

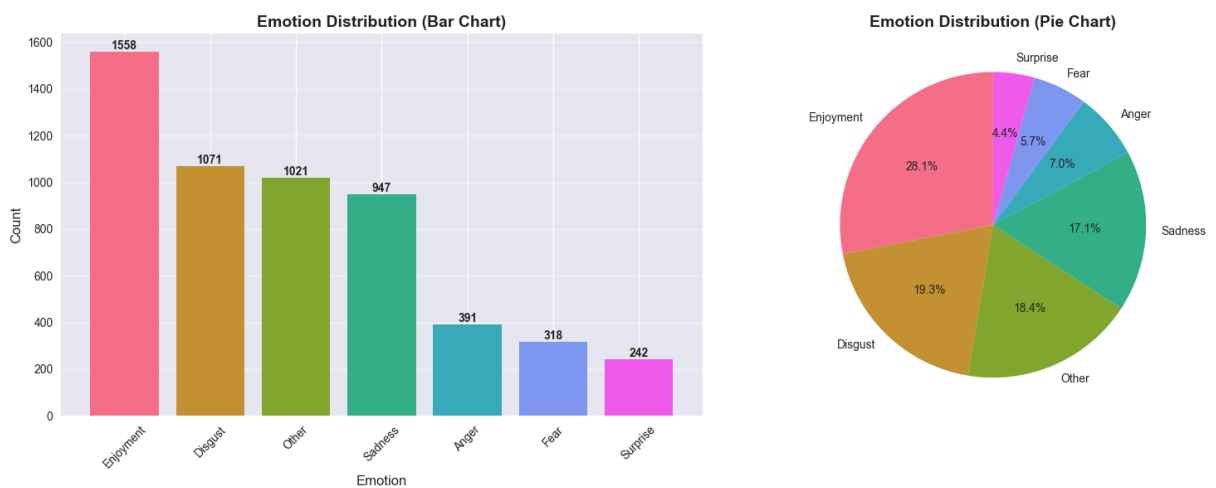


Figure 3.4 Data Distribution of Dataset.

Although the imbalance is not a favorable indicator for the dataset in ML, it reflects a real-life distribution, where the emotions are not equally distributed among the generated contents on the online platforms.

The database is designed to have 2 columns: text and label. The text column is the generated content collected from various sources. And the label is the annotated emotion for the sentence. The example data is shown below in Table 3.4.

Table 3.4 Example of Emotion-based Sentences.

No.	Vietnamese Sentences	English translation	Emotion
1	mỗi lần có video của con là cứ coi đi coi lại hoài . cưng con quá .	I can't stop watching videos of the little one. Such a cutie!	Enjoyment
2	nếu thấy phụ nữ quá phức tạp để hiểu và chinh phục thì đây là cách giải quyết vấn đề	Finding women impossible to understand or attract? Here's the solution.	Other
3	thời buổi bây chừ chuyện gì cũng có thể xảy ra , vậy nên thôi trất , đừng nên tin ai	Anything can happen these days, so honestly, it's best not to trust anyone.	Fear
4	xin đi chơi với bạn ngày tổng kết không cho cứ bắt học miết :)) khá mệt	Tried to go out for our end-of-year hangout but got shut down. All they want me to do is study... It's exhausting. :))	Sadness
5	đéo được tích sự gì 😂	Freaking useless. 😂	Anger
6	chủ yếu là bọn trẻ đú với bọn dân trí thấp mà thích thể hiện bác nhĩ	Mainly just posers and low-class people trying to show off, don't you think?	Disgust
7	canh thời gian đăng đúng vãi !	Talk about perfect timing!	Surprise

Table 3.5 Average Text Length by Emotion in The Original UIT-VSMEC.

Emotion	Avg Characters	Avg Words
Anger	72.15	18.34
Disgust	68.28	17.11
Fear	58.82	14.88
Sadness	56.04	13.98
Enjoyment	50.64	12.55
Other	48.84	12.2
Surprise	36.59	9.29

The average sentence length ranges from approximately 9 to 18 words across all emotional categories. The emotion that has the longest average words per content is anger, with an average of around 18 words. The second highest average is for Disgust, which has approximately 17 words. These findings suggest that negative feelings usually require more effort to express than other emotions. Meanwhile, surprise demonstrates the lowest average words per content. This reflects the reality of the collected data, since humans tend to use shorter words and reactionary exclamations when facing an unexpected event. This range is consistent with the nature of social media data, which predominantly consists of brief comments and short posts.

3.4. Adaptation Study with UIT- VSFC

3.4.1. Motivation

After training and evaluating our method with the UIT-VSMEC dataset, another experiment was set up to verify the ability of adaptation in a near-match task of our proposed framework.

The proposed stacked ensemble learning and LLM were trained and tested on the UIT-VSFC dataset. This experiment specifically highlights the distributional gap between our primary research domain and the UIT-VSFC dataset. The UIT-VSMEC corpus comprises informal social media text labeled with seven distinct, fine-grained emotion categories. In contrast, the UIT-VSFC dataset represents a more formal domain, consisting of university student feedback categorized into a three-class sentiment polarity (positive, neutral, negative). We frame this experiment as an adaptation test rather than a standard benchmark comparison.

The objective is to determine how well our ensemble pipeline transfers to a new domain when subjected to a significant reduction in label space. By applying the stacked ensemble learning framework to the sentiment-focused VSFC dataset, we aim to measure the stability of the ensemble learning. This assessment provides critical insights into whether the contextual reasoning captured by our model is truly generalizable or if it is constrained by the specific emotion-label definitions of the original corpus.

3.4.2. Methodology

To evaluate the adaptability of our proposed framework, we applied the stacked ensemble learning method to the UIT-VSFC dataset. Structurally, this adaptation mirrors the pipeline developed for the VSMEC corpus, utilizing the same five-base-learner architecture followed by accuracy-weighted fusion and the QLoRA Gemma meta-model. Importantly, all base models were retrained specifically on the UIT-VSFC data; to ensure an unbiased assessment, no VSMEC-derived artifacts or pre-trained parameters were utilized in this process.

The data preprocessing pipeline for the UIT-VSFC adaptation study strictly mirrors the methodology established for the VSMEC corpus to maintain experimental consistency. Each sentence undergoes a sequential cleaning process, including the replacement of Unicode emojis with descriptive Vietnamese words using a 116-item dictionary, the conversion of common text emoticons, and the expansion of Vietnamese internet abbreviations. Special characters, URLs, and HTML tags are removed, while punctuation is normalized and excess whitespace is stripped. Crucially, stopwords removal is avoided to preserve critical sentiment signals for the BERT-based models. Following these cleaning steps, deduplication is performed within each data split to ensure the integrity of the original partition. Finally, the training set was shuffled using a fixed random state (`random_state=42`) to ensure reproducibility across all experimental runs.

Subsequent to the preprocessing phase, the data is utilized to train five base learners, which includes PhoBERT, CafeBERT, ViBERT, Logistic Regression, and LinearSVC, consistent with the primary ensemble pipeline. Each base model is subjected to a 3-fold cross-validation process. During this stage, we extract the class-probability predictions from the learners to construct the new dataset. This novel data, which summarizes model confidence and weighted-average agreement, serves as the training data for the meta-LLM-model, specifically the QLoRA-adapted Gemma-2-9B. Finally, the predictive performance and stability of this novel ensemble model were rigorously tested on the UIT-VSFC test data to evaluate its cross-domain adaptability.

3.4.3. Dataset Overview

Table 3.6 UIT-VSFC Example Data.

Sentence	Sentence in english	Label
giảng viên nhiệt tình trong công tác giảng dạy	The lecturer is enthusiastic about their teaching	Positive
cách mà cô tiếp cận với sinh viên	The way she engages with students	Neutral
giáo viên không giảng dạy kiến thức , hướng dẫn thực hành trong quá trình học	The teacher does not teach the course content, only guiding practical work during the learning process.	Negative

The Vietnamese Students' Feedback Corpus (UIT-VSFC) serves as a significant linguistic resource, comprising over 16,000 sentences meticulously human-annotated for two distinct natural language processing tasks: sentiment-based and topic-based classifications. Within the scope of this research, we focus exclusively on the sentiment identification task. This specific focus is driven by the objective to validate our proposed framework against a domain-relevant task that mirrors the characteristics of the primary UIT-VSMEC dataset, allowing for a rigorous assessment of the model's adaptability and stability across different Vietnamese text corpora. Unlike the informal social media expressions in VSMEC, the VSFC dataset provides a more formal context of university student feedback, categorized into a three-class sentiment polarity: positive, neutral, and negative.

By applying our stacked ensemble learning pipeline to this corpus, we aim to gain critical insights into whether the contextual reasoning and predictive capabilities developed are truly generalizable or remain constrained by the specific emotional label definitions of the original social media domain.

CHAPTER 4

EXPERIMENTS AND RESULTS

4.1. Experiments

The first section investigates whether large language model (LLM)-based data augmentation can mitigate class imbalance in the UIT-VSMEC Vietnamese emotion corpus and improve emotion classification performance. The focus is on underrepresented emotions, particularly fear, anger, and surprise, which originally have fewer examples than other categories in UIT-VSMEC. Thereafter, the second part experiments with the BERT models, machine learning, and deep learning models in order to retrieve the 5 best models for the Vietnamese emotion recognition task. Subsequently, the predictions and probabilities for each of them are collected to create a new enhanced dataset for meta-model training later.

4.1.1. Augmentation

Two augmentation strategies using Gemini-2.5-flash were applied. First, a 10-shot prompting scheme (Appendix A) was used to generate synthetic examples for minority classes, with the goal of adding diverse yet emotionally consistent samples; semantic content may differ from the original sentences, but the emotion label is preserved. Second, paraphrasing prompts (Appendix B) were employed to rewrite existing sentences while maintaining their core semantic meaning and emotion label, thereby providing multiple surface realizations for the same emotion-bearing content. Both strategies are expected to enhance model generalization and lead to higher macro F1-scores in downstream evaluation.

The setting of Gemini-2.5-flash are:

- temperature=0.9 - high temperature encourage diversity [11]
- max_concurrency=50
- freq_penalty=0.5
- verbose=True

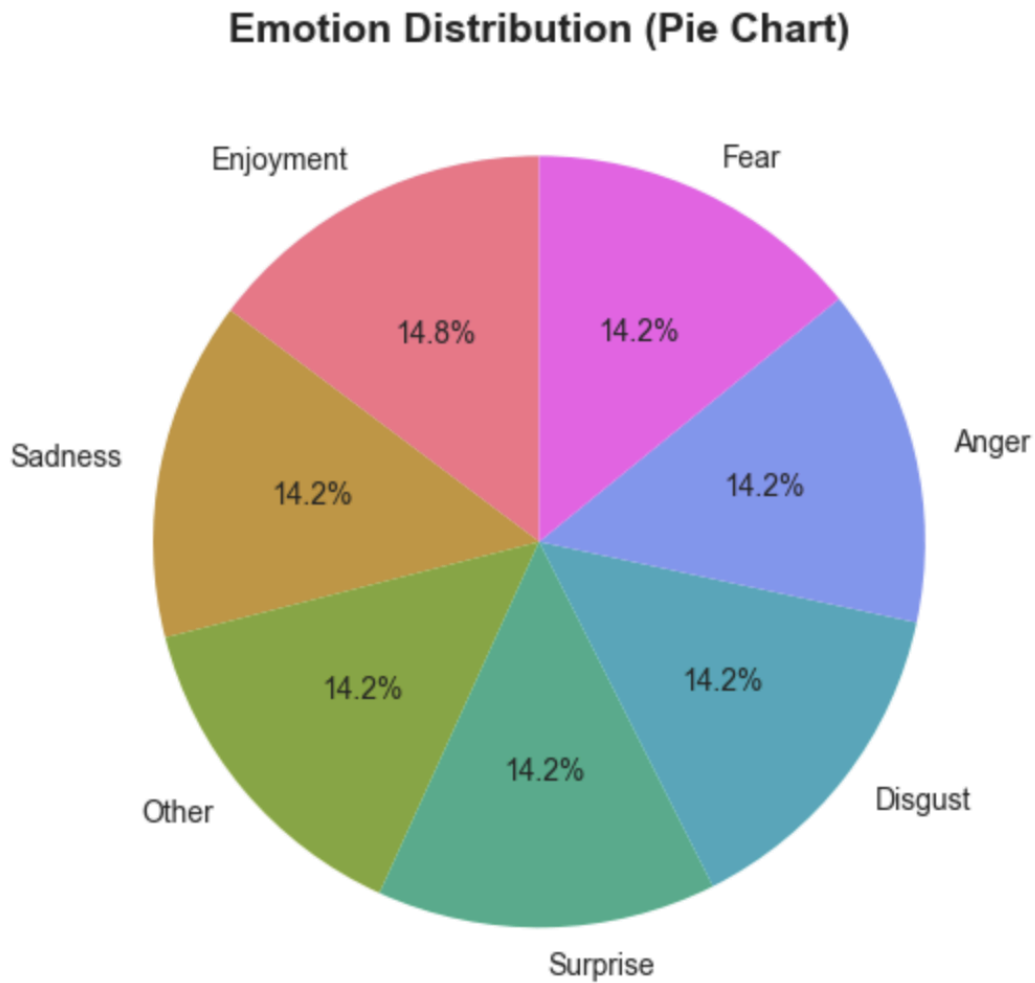


Figure 4.1 Emotion Distribution in The Augmented Dataset.

After augmentation, the dataset becomes equally distributed. This solves the problem of imbalance classes in the dataset.

4.1.2. Models & Baseline

To establish a baseline result for the testing and improvement of prompt data augmentation, several Vietnamese monolingual BERT models and one Roberta model were trained and evaluated using the original dataset. BERT models were chosen as the primary model type because they have consistently demonstrated significantly better performance than traditional machine learning models. BERT-based models have shown strong performance on Vietnamese NLP tasks. For instance, on the UIT-VSFC dataset, traditional supervised classifiers such as Naive Bayes and Maximum Entropy achieved F1-scores of approximately

87.5%, while deep learning methods achieved 92.0%. Similarly, PhoBERT has demonstrated superior performance on Vietnamese POS tagging (96.7-96.8% accuracy) and NER (93.6-94.7% F1-score), significantly outperforming traditional approaches[12]. For complex tasks like fine-grained emotion recognition, early neural network models such as CNN combined with word2vec achieved a weighted F1-score of only 59.74%[3]. The models tested included PhoBERT-Large v2, ViBERT, CafeBERT, and XML-Roberta.

The original data, after processing, was input into various BERT models. The model was then trained and tested using the original test dataset five times. Five different seeds were used to ensure that the results are fair and reproducible. The average of the five runs serves as the final metric for comparison. The best performer became the baseline result for this study.

4.1.3. Evaluation Setup

The assessment of model performance was conducted using two primary metrics: Accuracy and weighted F1-score, both of which are recognized as standard benchmarks for multi-class text classification and fine-grained emotion recognition. Accuracy provides a straightforward measure of total correct predictions across all emotion categories; however, it remains highly sensitive to the severe class imbalance present in the UIT-VSMEC dataset, where dominant labels like "Enjoyment" can mask poor performance in minority classes.

To provide a more robust evaluation, the weighted F1-score was utilized, as it accounts for label frequency by calculating the F1-score for each emotion category and weighting them by their respective support in the dataset. This dual-metric approach ensures that the model's effectiveness is captured both in its overall predictive capability and its ability to generalize across underrepresented emotional states like Fear, Anger, and Surprise.

4.1.4. Augmented Data Experimental Setup

To perform the model training in this phase, this study used three different training datasets and two different test sets. The three training datasets are the original data, augmented data created by paraphrase, and augmented data consisting of newly generated data. To evaluate the original training data's performance on real-world problems, a test set was created using an LLM with a new data-generating method. Subsequently, all the datasets were processed through the same data cleaning pipeline, which included replacing emojis with their corresponding descriptive words, replacing common abbreviations, and removing unnecessary whitespaces and special characters.

Once the cleaning process was finished, the experiments began. The first experiment involved training the BERT model with the original data and testing it on the generated test set. This experiment aimed to evaluate the model's robustness when trained only with the original data. The second experiment used the paraphrased method dataset for model training, followed by testing on the original dataset and the generated dataset consecutively. The third experiment involved training with the newly generated dataset, followed by testing with the original and generated test sets. The objective of these experiments was to evaluate the improvement in model performance achieved by the augmented data.

4.1.5. Ensemble Learning Experimental Setup

The research implements fine-grained Vietnamese emotion classification using a stacked ensemble architecture, leveraging established train, validation, and test splits alongside a standardized label mapping.

The foundational layer of the framework incorporates five heterogeneous base learners: PhoBERT, CafeBERT, ViBERT, Logistic Regression, and LinearSVC. Each individual base model undergoes training via 3-fold stratified prediction, followed by a refitting process on the comprehensive training set to derive probability vectors for the validation and testing phases.

Stacking weights are derived from validation accuracy metrics, and the meta-representation integrates the original textual context, per-model confidence distributions, and weighted-average probability signals. A Gemma-2-9B-it meta-model, adapted with QLoRA, is fine-tuned on this structured meta-corpus using completion-only supervision to predict final emotional labels. To mitigate information leakage and improve generalization, the training meta-samples utilize OOF probabilities, whereas validation and test samples are processed using full-train predictions.

The overall system performance is evaluated against the weighted-average baseline, the best-performing single classifier, and zero-shot Gemma inference, utilizing metrics such as accuracy, Macro F1, Weighted F1, and detailed confusion matrices.

4.2. Experiment Results

4.2.1. Baseline BERT Models

Table 4.1 Accuracy and Weighted F1 for Different BERT Models.

Model	Accuracy	Weighted F1
PhoBERT v2 -Large	0.6075	0.602
ViBERT	0.6106	0.6101
CafeBERT	0.6344	0.628
XML-Roberta	0.5691	0.5574

The table shows that all the Vietnamese monolingual BERT models outperform XML-Roberta. CafeBERT achieved the best result, with an accuracy of 0.6344 and a weighted F1 of 0.628. ViBERT (VisoBERT) followed closely behind with a minor gap, achieving an accuracy of 0.6106 and a weighted F1 of 0.6101. Although PhoBERT is well-known for its strong performance in Vietnamese NLP tasks, it appears to be underperforming in this particular task. Consequently, CafeBERT is selected as the main model for evaluating and testing the data augmentation method with Gemini-2.5-flash.

4.2.2. Data Augmentation with Gemini-2.5-flash

Table 4.2 F1-Scores for Minority Emotion Classes (Anger, Fear, Surprise) by Augmentation Strategy with CafeBERT.

	Fear	Anger	Surprise
Original	67%	38%	60%
Paraphrased	67%	42%	61%
Generated	68%	46%	61%

Table 4.2 shows that both augmentation strategies lead to steady improvements in all three weak emotion classes. The generated-data approach produces the most significant improvements, particularly in the anger class, where the F1-score rises from 38% in the original configuration to 46%. Improvements for fear and surprise are more modest but remain stable, indicating that data augmentation primarily benefits classes with higher initial imbalance or ambiguity. In general, these results show that augmentation works to improve

the model's ability to recognize emotions that aren't well represented. Generated data gives better performance gains than paraphrased data.

4.2.3. Model Robustness Testing

To test the model's robustness, a new test dataset was generated using the same methodology as the previously mentioned generated dataset. Subsequently, three datasets, which consists of original, paraphrased, and generated, were utilized to train the CafeBERT models. The results are detailed in Table 4.3.

Table 4.3 Accuracy, Macro F1, Weighted F1 of CafeBERT.

Data	Accuracy	Weighted F1
Original	0.6344	0.628
Paraphrased	0.6676	0.6682
Generated	0.6703	0.671

These findings indicate that increasing the training data through augmentation substantially improves model performance and generalization. When trained solely on the original dataset, the CafeBERT model achieved an accuracy of 63.44% and a weighted F1 score of 62.8%. In contrast, incorporating augmented data during training led to marked performance gains, with the paraphrased data yielding 66.76% accuracy and a 66.82% weighted F1 score, and the newly generated data achieving the highest results at 67.03% accuracy and a 67.1% weighted F1 score. These results demonstrate the effectiveness of LLM-based data augmentation in enhancing overall classification performance.

4.2.4. Models Overview

The study evaluates two broad groups of models for Vietnamese fine-grained emotion recognition on the UIT-VSMEC corpus: conventional machine learning and sequence models from the model running sheet, and transformer-based language models from the pre-thesis draft. The conventional group includes Logistic Regression, Support Vector Machine, Naive Bayes, and Random Forest using TF-IDF vectorization, together with CNN and Bi-LSTM using tokenization and padding; the transformer group includes PhoBERT-Large v2, VisoBERT, CafeBERT, and XLM-RoBERTa.

From a methodological perspective, the TF-IDF-based models rely on sparse lexical representations and are effective when label discrimination is strongly associated with word frequency patterns, whereas CNN and Bi-LSTM attempt to learn sequential and local contextual patterns from tokenized text. By contrast, the BERT-family models in the thesis exploit contextualized embeddings learned from large corpora, which explains why they were selected as the main baseline family for the core experiments.

4.2.4.1. Results

Table 4.4 Model Comparison Train on Original Data.

Model	Vectorization	Test Accuracy	Test F1
Logistic Regression	TF-IDF	0.5695	0.5751
SVM (LinearSVC)	TF-IDF	0.5657	0.5592
Random Forest	TF-IDF	0.5203	0.5535
Naive Bayes	TF-IDF	0.4769	0.4159
CNN	Tokenization + Padding	0.5578	0.5309
Bi-LSTM	Tokenization + Padding	0.4682	0.4118

Table 4.5 Model Comparison Train on Augmented Data.

Model	Vectorization	Test Accuracy	Test F1
Logistic Regression	TF-IDF	0.5679	0.5721
SVM (LinearSVC)	TF-IDF	0.5621	0.5653
Random Forest	TF-IDF	0.5376	0.5441
Naive Bayes	TF-IDF	0.5246	0.5312

CNN	Tokenization	+		
	Padding		0.422	0.416
Bi-LSTM	Tokenization	+		
	Padding		0.3035	0.2511

The comparative analysis of non-transformer models across both the original and augmented datasets (Tables 4.4 and 4.5) establishes a clear and consistent trend. When trained solely on the original data (Table 4.4), the linear TF-IDF models exhibited superior stability and performance. Logistic Regression recorded the highest metrics, achieving a Test F1 of 0.5751 and a Test Accuracy of 0.5695, closely followed by SVM (LinearSVC) at 0.5592 Test F1 and 0.5657 Accuracy. In contrast, the deep learning architectures showed significant weakness, with CNN (Test F1: 0.5309) and Bi-LSTM (Test F1: 0.4118) posting the lowest classification scores.

Crucially, the relative model hierarchy was maintained when training utilized the augmented dataset (Table 4.5). Logistic Regression and SVM retained their status as the top performers, yielding Test F1 scores of 0.5721 (Accuracy: 0.5679) and 0.5653 (Accuracy: 0.5621), respectively. Significantly, the neural sequence models demonstrated a marked decline in effectiveness under the augmented condition, with CNN's Test F1 dropping to 0.416 and Bi-LSTM's Test F1 severely falling to 0.2511. This decline in neural sequence model performance suggests that their architectural limitations were exceeded by the increase in data complexity and volume introduced by LLM augmentation, which ultimately made linear TF-IDF representations more robust by comparison.

Table 4.6 BERT Model with Augmented Comparison.

Model	Test Accuracy	Test F1
CafeBERT	0.6703	0.671
PhoBERT	0.6378	0.633
ViBERT	0.6387	0.6375
XML-R	0.6214	0.6227

The transformer baselines reported were consistently stronger than the non-transformer baselines on the original test set. As shown in Table 4.6, CafeBERT achieved

the best transformer baseline with a Test Accuracy of 0.6703 and a Test F1 score of 0.671. ViBERT and PhoBERT followed closely, with ViBERT recording a Test Accuracy of 0.6387 (Test F1: 0.6375) and PhoBERT registering a Test Accuracy of 0.6378 (Test F1: 0.633). XLM-RoBERTa consistently yielded the lowest metrics among the BERT-family models, with a Test F1 of 0.6227. These strong augmentation results paved the way for the full LLM-Ensemble framework, which ultimately achieved the highest overall performance in this research, recording 70.23% accuracy and a 70.22% weighted F1 score on the VSMEC test set.

4.2.5. Ensemble Learning Result - UIT VSMEC

Table 4.7 Performance Comparison between Ensemble Learning Methods and Others with UIT-VSMEC.

Name	Accuracy	F1_weighted	F1_macro
Our proposed framework	0.702312	0.702217	0.6831
Gemma + zero shot prompting	0.686416	0.680933	-
MLR+ pre-processing techniques + key-clause extraction [17]	0.6436	0.6440	-
CafeBERT[16]	-	0.6612	-
VisoBERT[21]	0.6810	0.6837	-
XLM-R-Large[21]	0.6137	0.60202	-
GRU + CNN + BiLSTM + LSTM [23]	-	-	0.6579

The comprehensive evaluation results for various classification models on the VSMEC dataset are detailed in Table 4.7. The experimental data reveals that the proposed LLM-Ensemble framework demonstrates superior performance, effectively outperforming all other baseline and state-of-the-art models by achieving a peak accuracy of 0.7023 and a weighted F1 score of 0.7022. This result signifies a notable advancement over the second-best performing configuration, which is the zero-shot prompting approach with Gemma reaching a weighted F1 score of 0.6809. Furthermore, the proposed method consistently exceeds the

capabilities of standalone monolingual Vietnamese BERT models, such as CafeBERT (0.6612) and VisoBERT (0.6837), as well as deep learning approaches like the GRU + CNN + BiLSTM + LSTM ensemble (0.6579 F1-macro). These findings validate that the integration of diverse base learners with an LLM meta-model effectively addresses the complexities of fine-grained emotion recognition in low-resource settings.

4.2.6. Ensemble Learning Result - UIT VSFC

Table 4.8 Performance Comparison between Ensemble Learning Methods and Others with UIT-VSFC.

Model	Accuracy	Recall	Macro F1	Weighted F1
PhoBert(Base) + CNN-LSTM [15]	93.24	93.24	92.92	-
CafeBERT [16]	-	-	84.29	-
MaxEnt classifier [5]	-	88.66	87.94	-
Feature fusion - Two best models [7]	-	-	94.00	93.76
Our Ensemble Learning + LLM (Gemma)	94.13	81.17	83.51	93.84
(BERT + CNN + BiLSTM + LSTM) [23]	-	-	92.79	-

The comprehensive evaluation outcomes for various classification systems on the UIT-VSFC corpus are presented in Table 4.8. Experimental data indicates that the proposed LLM-Ensemble framework exhibits superior performance, reaching a peak accuracy of 94.13% and a 93.84% weighted F1 score. This performance constitutes a notable advancement over alternative configurations, including the PhoBERT + CNN-LSTM architecture [15] (93.24% accuracy) and the BERT + CNN + BiLSTM + LSTM ensemble [23] (92.79% macro-F1). Moreover, the proposed strategy consistently surpasses the capabilities of the feature fusion method (93.76% weighted F1) as well as the MaxEnt classifier [5] (87.94% Macro F1). These results confirm the adaptability of integrating

heterogeneous base learners with a generative LLM meta-model, illustrating its efficacy across diverse domains and label distributions.

CHAPTER 5

DISCUSSION AND EVALUATION

5.1. Discussion

The primary objective of this research is to first mitigate the severe class imbalance found in foundational Vietnamese emotion corpora like UIT-VSMEC, where dominant labels like "Enjoyment" (28.36%) often overshadow minority categories such as "Surprise" (4.46%) or "Anger" (6.92%). The findings demonstrate that leveraging Gemini-2.5-Flash for targeted data augmentation provides a robust solution to the data scarcity problem.

A key point of discussion is the superiority of generated data over paraphrased data. While both strategies improved model generalization, the approach of generating completely new synthetic instances resulted in higher macro F1-score gains compared to simple paraphrasing of existing human-annotated sentences. This aligns with broader surveys suggesting that LLM-generated data can, in some instances, exceed the value of human-curated data because it is not restricted by the linguistic patterns or noise present in the original small-scale dataset [1].

The second goal of this study is to test and evaluate a novel method for fine-grained emotion recognition tasks in Vietnamese. The proposed framework implements a stacked ensemble architecture that combines weighted average ensemble learning with the generative power of a Large Language Model (LLM). By utilizing ensemble learning to aggregate predictions from diverse base learners, such as CafeBERT and specialized machine learning models, the system aims to reduce individual model errors and variance. This is further enhanced by the contextual reasoning capability of the Gemma-2-9B meta-model, which is fine-tuned to interpret inter-model agreement and uncertainty patterns. The research specifically targets the nuances of Vietnamese social media text to improve classification reliability. The results indicate that this hybrid approach offers significant potential for improvement in fine-grained emotion recognition for Vietnamese natural language processing, as evidenced by the higher accuracy and weighted F1 scores achieved compared to standalone individual models.

The adaptation study conducted on the UIT-VSFC corpus provides important empirical evidence regarding the framework's ability to adapt across domains and its structural robustness. In contrast to the primary experiments on the UIT-VSMEC corpus,

which focused on fine-grained, informal emotion labels (7-class), the UIT-VSFC adaptation evaluated the framework's performance in a formal domain defined by 3-class sentiment polarity.

5.2. Comparison

The performance of the proposed LLM-Ensemble framework demonstrates a significant advancement over existing methods for fine-grained Vietnamese emotion recognition on the UIT-VSMEC dataset. The proposed framework, utilizing the LoRA fine-tuned Gemma-2-9B meta-model, achieved the highest overall results with an accuracy of 70.23% and a weighted F1 score of 70.22%. This performance surpasses all other configurations evaluated in this study, including the best-performing single model, CafeBERT, which achieved 66.76% accuracy and 66.82% weighted F1, as well as the zero-shot Gemma approach, which reached 68.64% accuracy and 68.09% weighted F1.

The proposed LLM-ensemble framework, achieving 70.23% accuracy and a 70.22% weighted F1, markedly surpasses the performance of the best Vietnamese monolingual BERT models, which recorded weighted F1 scores ranging from 66.82% (CafeBERT) to 68.37% (VisoBERT). This result also exceeds the performance of zero-shot Gemma inference (68.64% accuracy, 68.09% weighted F1) and offers a substantial improvement over the 59.74% weighted F1 score of the CNN+word2vec baseline. Furthermore, these findings outshine traditional machine learning approaches like Support Vector Machines (SVM) and Random Forest, which achieved weighted F1 scores of 56.87% and 40.11%, respectively, on the UIT-VSMEC dataset. This validates that integrating heterogeneous base learners with an LLM meta-model effectively leverages collective predictive power to overcome individual classifier limitations in low-resource NLP settings.

The results presented in Table 4.8 indicate that the stacked ensemble architecture maintains its superior predictive efficacy with the UIT-VSFC data. Achieving a peak accuracy of 94.13% and a weighted F1 score of 93.84%, the framework not only maintained high performance but also outperformed specialized benchmarks such as the PhoBERT + CNN-LSTM architecture [15] (93.24% accuracy) and the BERT + CNN + BiLSTM + LSTM ensemble [23] (92.79% Macro F1).

5.3. Evaluation

This study successfully demonstrated the effectiveness of an LLM-augmented data approach in addressing the limitations of fine-grained emotion recognition (ER) in Vietnamese, a low-resource NLP task.

The primary measure of success was the significant improvement in the model's robustness and macro F1 score, particularly for minority classes. The CafeBERT model trained on the newly generated synthetic data achieved an accuracy of 67.03% and a weighted F1 score of 67.1% on the test set. This represents a substantial performance gain compared to the 63.44% accuracy and 62.8% weighted F1 score achieved by the model trained only on the original, imbalanced dataset. Furthermore, this result far exceeds the 59.74% weighted F1-score of the previous state-of-the-art CNN+word2vec baseline on the UIT-VSMEC corpus.

The evaluation highlights two critical successes:

1. **Mitigation of Data Imbalance:** The targeted augmentation using Gemini-2.5-Flash successfully enriched the training data for underrepresented emotions (Fear, Anger, Surprise), evidenced by the notable increase in their respective F1-scores (e.g., Anger rising from 38% to 46%). This validated the first objective of leveraging LLMs for controlled data augmentation.
2. **Enhanced Model Robustness:** The significant performance leap when testing on the newly generated test set (an increase from 62.8% to 67.1% in weighted F1) strongly indicates that the LLM-generated data improved the model's generalization capabilities beyond the patterns present in the original human-annotated corpus.

The experimental results also demonstrate the high efficacy of the proposed ensemble-LLM framework for fine-grained Vietnamese emotion recognition. As evidenced by the data presented in Table 4.7, the proposed method—utilizing the LoRA fine-tuned Gemma meta-model—attained the highest level of performance, recording an accuracy of 70.23% and a weighted F1-score of 70.22%. This represents a notable advancement over all baseline configurations, including the highest-performing monolingual classifier, CafeBERT (66.76% accuracy, 66.82% weighted F1), and the zero-shot prompting approach with Gemma (68.64% accuracy, 68.09% weighted F1).

The performance gain highlights the advantage of integrating heterogeneous base learners with a large language model. The weighted averaging mechanism effectively aggregates diverse predictive signals from multiple models, reducing individual model biases

and variance. At the same time, the LLM meta-model leverages its contextual reasoning capability to interpret the structured meta-representation, capturing complex relationships between model confidences and input semantics. This combination enables more robust and accurate predictions compared to standalone models.

Moreover, the use of 3 fold cross validation predictions ensures that the meta-model is trained on unbiased probability estimates, mitigating information leakage and improving generalization. The results also indicate that while zero-shot LLMs already provide competitive performance, fine-tuning with ensemble-informed features yields further improvements, confirming the benefit of incorporating task-specific signals.

Furthermore, the success in the UIT-VSFC experiment is particularly significant for two reasons. First, it demonstrates that the benefits of the LLM-Ensemble architecture are not restricted to the specific label space of fine-grained emotion recognition. Second, it confirms that the meta-model’s ability to interpret inter-model agreement and uncertainty is generalizable. By successfully adapting to a 3-class sentiment task without requiring modifications to the core ensemble pipeline, the framework proves its capability to handle varying degrees of granularity and complexity. Consequently, these findings validate that the proposed LLM-Ensemble architecture is a flexible and reliable solution for diverse Vietnamese NLP tasks, effectively decoupling the ensemble mechanism from the specific emotional definitions of the original corpus.

However, despite achieving the best overall performance, the improvement margin remains moderate, suggesting that fine-grained emotion recognition in Vietnamese continues to be a challenging task. This may be attributed to linguistic ambiguity, subtle emotional expressions, and dataset limitations. Therefore, future work could explore richer feature representations, more diverse base learners, and advanced prompting or alignment strategies to further enhance performance.

CHAPTER 6

CONCLUSION AND FUTURE WORK

6.1. Conclusion

The results of this research demonstrate the high efficacy of an LLM-augmented approach for enhancing fine-grained Emotion Recognition (ER) in Vietnamese, a characteristic low-resource language. Targeted data augmentation utilizing Gemini-2.5-Flash successfully mitigated the foundational challenges of data scarcity and severe class imbalance within the UIT-VSMEC corpus. The primary finding is the substantial improvement in classification performance achieved through synthetic data generation; this performance represents a significant leap over models trained on original data and dramatically exceeds the previous CNN+word2vec Weighted F1 score of 59.74%.

Furthermore, this study validated the proposed LLM-Ensemble framework as the most effective architecture for this task. The framework integrating heterogeneous base learners with a LoRA fine-tuned Gemma meta-model attained the highest overall level of performance on the VSMEC dataset, recording an accuracy of 70.23% and a Weighted F1 score of 70.22%. This result surpasses the highest-performing baseline monolingual classifier, CafeBERT, which achieved a Weighted F1 score of 66.82%.

Beyond the primary emotion recognition task, the research further validated the proposed LLM-Ensemble framework's cross-domain adaptability through an adaptation test on the UIT-VSFC dataset. Demonstrating its structural flexibility, the framework achieved a peak accuracy of 94.13% and a 93.84% weighted F1 score, successfully outperforming existing benchmarks such as the PhoBERT + CNN-LSTM architecture and BERT-based ensemble models. This successful transition from fine-grained emotion recognition (7-class) to sentiment polarity classification (3-class) confirms that the ensemble mechanism effectively decouples from specific label definitions, proving its reliability as a flexible solution across diverse Vietnamese NLP tasks.

In summary, this research confirms that Large Language Models serve as powerful tools for creating high-quality synthetic data and robust ensemble frameworks, offering a viable solution for advancing Vietnamese NLP research where traditional annotated data is scarce.

6.2. Future Work

The current study established the effectiveness of LLM-based data augmentation for fine-grained Vietnamese Emotion Recognition (ER). Building upon these results, future research should focus on the following key areas:

1. Cross-dataset generalization: While an adaptation test was implemented, the current evaluation focused primarily on sentiment classification. Future work should involve additional experiments exploring other facets of natural language to further verify the model's generalizability and capability across a wider array of NLP tasks.
2. Data Validation and Quality Assurance Pipeline: Develop a dedicated pipeline for systematically validating the quality and emotional consistency of the LLM-generated synthetic data. This could involve human-in-the-loop review of a subset of generated examples or automated metrics to ensure that the augmented data maintains the linguistic and emotional characteristics of real social media text, thereby sustaining the model's generalization capability.
3. Comparative Analysis of Augmentation Techniques: Conduct a deeper comparison between various LLM prompting strategies (e.g., zero-shot, few-shot, chain-of-thought) and different LLMs to identify the most cost-effective and highest-quality data augmentation method for low-resource ER.
4. Improved weight learning: Instead of using fixed validation-accuracy-based weights, subsequent research should investigate dynamic weighting strategies, such as sample-level gating networks or uncertainty-aware weighting. These methods may better exploit complementary strengths of base models and improve performance on difficult or ambiguous samples.

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APPENDICES

Appendix A

"system", ""You are an expert in Vietnamese language, social media writing style.

Your task is to generate Vietnamese sentences that are related to the given emotion while STRICTLY following the rules below:

- All outputs must sound natural, realistic, and suitable for Vietnamese social media.
- Only generate sentences that express the given emotion.
- Use slang or emoji sparingly and only when they feel natural.

""),

("human", ""Generate exactly {num_variations} NEW Vietnamese sentences that express the given emotion as the example sentences below.

Emotion to generate:

- English label: "{emotion}"
- Vietnamese description: "{emotion_vn}"

Example sentences:

1. "{example1}"
2. "{example2}"
3. "{example3}"
4. "{example4}"
5. "{example5}"

6. "{example6}"
7. "{example7}"
8. "{example8}"
9. "{example9}"
10. "{example10}"

STYLE & LENGTH CONSTRAINTS:

5. Length variation is required across outputs:
 - Short: 5–10 words
 - Medium: 10–20 words
 - Long: 20+ words
6. All sentences must be in Vietnamese.
7. Your sentence must not contain any other emotion than the given emotion.

OUTPUT FORMAT (STRICT):

Each line must follow this exact format:

[number]. [new Vietnamese sentence] (example [source_number])

Where:

- [number] is 1 to {num_variations}
- [source_number] is 1 to 10 (indicating which example sentence was paraphrased)

Example output format:

1. câu tiếng việt mới (example 1)
2. một câu khác (example 7)

Rules:

- One sentence per line
- No explanations, notes, or extra text
- Always include (example X) at the end where X is 1 to 10

""

Appendix B

"system", ""You are an expert in Vietnamese Gen Z language, social media writing style.

Your task is to generate Vietnamese sentences while STRICTLY preserving:

- Core semantic meaning
- Emotional tone and intensity
- Speaker intent

All outputs must sound natural, realistic, and suitable for Vietnamese social media.

Do NOT introduce new facts, entities, time references, causes, or consequences.

Do NOT change sentiment polarity or emotional strength.

Use slang or emoji sparingly and only when they feel natural.

- DO NOT REPLACE: emotional particles (nhé, à, mà), negations (không, chẳng)
- ONLY augment: content words, not structure words
- Preserve: emotional intensifiers (rất, lắm, quá)

""),

("human", ""Generate exactly {num_variations} NEW Vietnamese sentences that preserve the SAME meaning and emotion as the example sentences below.

Emotion to preserve:

- English label: "{emotion}"
- Vietnamese description: "{emotion_vn}"

Example sentences:

1. "{example1}"
2. "{example2}"
3. "{example3}"
4. "{example4}"
5. "{example5}"
6. "{example6}"
7. "{example7}"
8. "{example8}"
9. "{example9}"
10. "{example10}"

SEMANTIC SIMILARITY CONSTRAINTS (MANDATORY):

1. Each generated sentence must be semantically equivalent to its source example:
 - No addition or removal of key actions, intentions, or outcomes
2. Preserve:
 - Subject (implicit or explicit)
 - Action/state
 - Target/object (if any)
 - Cause–effect relationship (if present)
3. Allowed changes:
 - Word choice (synonyms, paraphrasing)
 - Sentence structure (active/passive, reordering)
 - Discourse markers or softeners common in Gen Z Vietnamese
4. Forbidden changes:
 - Changing who is speaking or being referenced
 - Adding sarcasm if not present in the original
 - Adding intensity (e.g., “rất”, “quá trời”) unless present originally
 - Introducing new emotions or mixed emotions

STYLE & LENGTH CONSTRAINTS:

5. Length variation is required across outputs:

- Short: 5–10 words
- Medium: 10–20 words
- Long: 20+ words

6. All sentences must be in Vietnamese.

7. Gen Z style must be natural, not exaggerated.

8. Emojis or slang:

- Optional
- Maximum 1 emoji per sentence
- Must not change semantic meaning or emotion

OUTPUT FORMAT (STRICT):

Each line must follow this exact format:

[number]. [new Vietnamese sentence] (example [source_number])

Where:

- [number] is 1 to {num_variations}
- [source_number] is 1 to 10 (indicating which example sentence was paraphrased)

Example output format:

1. câu tiếng việt mới (example 1)
2. một câu khác (example 7)

Rules:

- One sentence per line
- No explanations, notes, or extra text
- Always include (example X) at the end where X is 1 to 10

Appendix C

C.1. Emotion mapping list

'😊': 'vui vẻ', '😄': 'vui vẻ', '😁': 'vui vẻ', '😆': 'vui vẻ', '😅': 'vui vẻ',
'😂': 'vui vẻ', '😜': 'cười lớn', '😝': 'cười lớn', '😏': 'mỉm cười', '😌': 'mỉm cười',
'😍': 'nháy mắt', '😄': 'hạnh phúc', '😇': 'thiên thần', '😘': 'yêu thương',
'😍': 'yêu thích', '😄': 'ngưỡng mộ', '😘': 'hôn', '😘': 'hôn', '😊': 'hạnh phúc',
'😘': 'hôn', '😘': 'hôn', '😘': 'vui buồn lẫn lộn',
'❤️': 'yêu thương', '💕': 'yêu thương', '💕': 'yêu thương', '💕': 'yêu thương',
'💕': 'yêu thương', '💕': 'yêu thương', '💕': 'yêu thương', '💕': 'yêu thương',
'💕': 'yêu thương', '💕': 'yêu thương', '💕': 'yêu thương', '💕': 'yêu thương',
'💕': 'yêu thương', '💕': 'yêu thương', '💕': 'yêu thương', '💕': 'yêu thương',
'😭': 'buồn bã', '😭': 'khóc', '😭': 'buồn bã', '😭': 'thất vọng',
'😭': 'buồn bã', '😭': 'lo lắng', '😭': 'bối rối', '😭': 'buồn bã',
'😭': 'buồn bã', '😭': 'đau khổ', '😭': 'khó chịu', '😭': 'mệt mỏi',
'😭': 'mệt mỏi', '😭': 'van xin', '😭': 'buồn ngủ', '😭': 'lo lắng',
'😡': 'tức giận', '😡': 'tức giận', '😡': 'tức giận', '😡': 'bực tức',
'😡': 'tức giận', '😡': 'tức giận', '😡': 'tình nghịch',
'😨': 'sợ hãi', '😨': 'lo sợ', '😨': 'kinh hãi', '😨': 'lạnh',
'😨': 'ngạc nhiên', '😨': 'ngạc nhiên', '😨': 'sửng sốt', '😨': 'xấu hổ',
'😨': 'choáng váng', '😨': 'căng thẳng', '😨': 'sợ hãi',
'😨': 'buồn nôn', '😨': 'ghê tởm', '😨': 'bệnh', '😨': 'hất xì',
'😨': 'ôm', '😨': 'đau', '😨': 'chóng mặt', '😨': 'say',
'😨': 'suy nghĩ', '😨': 'nghỉ ngơi', '😨': 'bình thường', '😨': 'bình thường',
'😨': 'im lặng', '😨': 'chán', '😨': 'ranh mãnh', '😨': 'không hài lòng',
'😨': 'ôm', '😨': 'che miệng cười', '😨': 'im lặng', '😨': 'nói dối',
'😨': 'ngủ', '😨': 'ngủ', '😨': 'tốt', '😨': 'tệ', '😨': 'được',
'😨': 'chiến thắng', '😨': 'chúc may mắn', '😨': 'cầu nguyện',
'😨': 'mạnh mẽ', '😨': 'nóng bỏng', '😨': 'lấp lánh', '😨': 'ngôi sao',
'😨': 'hoàn hảo', '😨': 'ngầu', '😨': 'thông minh', '😨': 'ăn mừng',
'😨': 'ngon', '😨': 'thèm', '😨': 'tình nghịch', '😨': 'tình nghịch',
'😨': 'lè lưỡi', '😨': 'điên', '😨': 'nhìn', '😨': 'nhìn',

C.2. Emoticon mapping list

':): 'vui vẻ', ':-)': 'vui vẻ', '(: ': 'vui vẻ',

'D': 'cười lớn', ':-D': 'cười lớn',
 ':))': 'cười lớn', ':)))': 'cười lớn', ':))))': 'cười lớn',
 '==))': 'cười lớn', '==)))': 'cười lớn',
 ':3': 'dễ thương',
 ':(': 'buồn bã', ':-(': 'buồn bã', ":'(": 'khóc',
 ':(((': 'buồn bã', '(((((': 'rất buồn',
 ':P': 'tỉnh nghịch', ':-P': 'tỉnh nghịch', ':p': 'tỉnh nghịch',
 ':)': 'nháy mắt', ';-)': 'nháy mắt',
 ':O': 'ngạc nhiên', ':-O': 'ngạc nhiên', ':o': 'ngạc nhiên',
 'o.O': 'ngạc nhiên', 'O.o': 'ngạc nhiên',
 '@@': 'choáng',
 '-_-': 'chán', '-.-': 'chán',
 '>.<': 'khó chịu',
 '^_^': 'vui vẻ', '^.^': 'vui vẻ',
 'T_T': 'khóc', 'T.T': 'khóc',
 '>_<': 'khó chịu',
 '<3': 'yêu thương', '</3': 'tan vỡ',
 ':*': 'hôn', ':-*': 'hôn',

C.3. Abbreviation mapping list

'ko': 'không', 'k': 'không', 'hok': 'không', 'hem': 'không', 'hông': 'không',
 'dc': 'được', 'đc': 'được', 'đk': 'được', 'dk': 'được', 'duoc': 'được',
 'vs': 'với', 'voi': 'với', 'v': 'với',
 'mk': 'mình', 'mik': 'mình', 'minh': 'mình',
 'bn': 'bạn', 'pn': 'bạn', 'bạng': 'bạn',
 'cx': 'cũng', 'cg': 'cũng', 'cug': 'cũng',
 'ntn': 'như thế nào', 'sn': 'sao',
 'bh': 'bây giờ', 'bjo': 'bây giờ', 'bjờ': 'bây giờ', 'bgiờ': 'bây giờ',
 'j': 'gì', 'jì': 'gì', 'z': 'vậy', 'zậy': 'vậy', 'r': 'rồi',
 'trc': 'trước', 'trog': 'trong', 'tg': 'thời gian',
 'nt': 'nhắn tin', 'nv': 'nhân vật',
 'ns': 'nói', 'nc': 'nói chuyện', 'nch': 'nói chuyện',
 'bt': 'bình thường', 'bth': 'bình thường', 'bình thuong': 'bình thường',

'thg': 'thằng', 'thag': 'thằng',
 'ng': 'người', 'ngta': 'người ta', 'nguoì': 'người',
 'ak': 'à', 'ak': 'ạ', 'nka': 'nhé', 'nhe': 'nhé',
 'gđ': 'gia đình', 'gd': 'gia đình',
 'hj': 'hihi', 'haha': 'haha', 'hihi': 'hihi', 'huhu': 'huhu',
 'kkk': 'haha', 'kk': 'haha', 'wkwk': 'haha',
 'vkl': 'rất', 'vl': 'rất', 'vcl': 'rất',
 'clgt': 'chắc luôn', 'cl': 'chắc luôn',
 'tl': 'trả lời', 'rep': 'trả lời',
 'acc': 'tài khoản', 'fb': 'facebook', 'ib': 'nhắn tin',
 'dt': 'điện thoại', 'dt': 'điện thoại',
 'tv': 'tivi', 'tp': 'thành phố', 'tphcm': 'thành phố hồ chí minh',
 'hn': 'hà nội', 'sg': 'sài gòn',
 'trg': 'trường', 'truong': 'trường',
 'lm': 'làm', 'lam': 'làm',
 'cty': 'công ty', 'cong ty': 'công ty',
 'thang': 'tháng',
 'qtr': 'quan trọng', 'qtrog': 'quan trọng',
 'sp': 'sản phẩm', 'dv': 'dịch vụ',
 'hq': 'hàn quốc', 'tq': 'trung quốc',
 'thik': 'thích', 'tk': 'tài khoản',
 't': 'tao',

Appendix D

[TEXT] ... [/TEXT]

[STACK]

<phobert w=0.205>{joy=0.910, sadness=0.040, ...}

<cafebert w=0.198>...

...

[/STACK]

[WEIGHTED_AVG]{joy=0.870, ...}[/WEIGHTED_AVG]

